

Better approximation to the susie-love model

1 TO-DO

Understand the relationship between

1. PIP only depends on x and the diagonal of R .
2. $p(x_{-j}|x_j, \beta, \gamma = j)$ does not depend on β .

Look Yuxin's thesis and read the SI of susie-rss paper more carefully.

Why does having β^2 in the variance make sense? Of course it follows from the IW or Wi model, but intuitively why does it make sense?

Is there another model that arises from the conditional distribution:

$$p(x_{-j} | x_j, \beta, \gamma = j) = \mathcal{N} \left(x_j \bar{R}_{j,-j}, \left(\frac{1}{N} + \frac{x_j^2}{\nu_0 + 3} \right) S_j \right)? \quad (1)$$

2 Some new models with better approximation to susie-love

Recall our original SER model from what is called 'susie-love':

$$\begin{cases} x|R, \beta, \gamma \sim \mathcal{N}(\beta R_\gamma, R/N) \\ R \sim \mathcal{IW}(\nu_0 \bar{R}, \nu_0 + J + 1). \end{cases} \quad (2)$$

Note that the latent matrix R appears in both mean and variance of the likelihood in (2). This model is very natural for unknown in-sample LD R , but is hard to make inferences. Last time we approximated R in the variance to be $\bar{R}$, which makes the inference much easier but might eventually lead to the loss of the "irrelevance of the null SNPs" property. In principle, it makes sense that when $L = 1$, PIP of SNP j calculation should only depend on x_j but not on other x_k and $\bar{R}_{jk}$ ("irrelevance of the null SNPs" property for SER).

I think this original model is actually feasible for inference. Consider the following equivalent way to represent it.

2.1 Model A (equivalent to the original SER-love model)

Let's write down $p(x|R, \beta, \gamma = 1)$, other cases of γ are similar. Let

$$d_1 = R_{11} \in \mathbb{R}_+, \quad z_1 = \frac{R_{-1,1}}{d_1} \in \mathbb{R}^{J-1},$$

and

$$S_1 = R_{-1,-1} - d_1 z_1 z_1^\top$$

be the Schur's complement of R_{11} in R . Hence, the latent R is represented as

$$R = \begin{pmatrix} d_1 & d_1 z_1^\top \\ d_1 z_1 & S_1 + d_1 z_1 z_1^\top \end{pmatrix}. \quad (3)$$

Similarly, denote

$$\bar{S}_1 = \bar{R}_{-1,-1} - \bar{R}_{-1,1} \bar{R}_{-1,1}^\top$$

be the Schur's complement of $\bar{R}_{11}$ in $\bar{R}$, and $\mu_1 = \bar{R}_{-1,1}$, we can write

$$\bar{R} = \begin{pmatrix} 1 & \mu_1^\top \\ \mu_1 & \bar{S}_1 + \mu_1 \mu_1^\top \end{pmatrix}. \quad (4)$$

Properties of IW distribution (together with the fact that $\bar{R}_{11} = 1$) imply, using the shape-scale parameterization for the inverse-gamma distribution,

$$\begin{cases} d_1 \sim \text{IG}\left(\frac{\nu_0 + 2}{2}, \frac{\nu_0}{2}\right) \\ z_1 | S_1 \sim \mathcal{N}\left(\bar{R}_{-1,1}, \frac{S_1}{\nu_0}\right) \\ S_1 \sim \text{IW}(\nu_0 \bar{S}_1, \nu_0 + J + 1) \end{cases} \quad (5)$$

The original SER version of Susie-love (with $\gamma = 1$; similar for other γ) becomes:

$$\begin{cases} x | \beta, \gamma = 1, z_1, S_1 \sim \mathcal{N}\left(\beta \begin{pmatrix} d_1 \\ d_1 z_1 \end{pmatrix}, \frac{1}{N} \begin{pmatrix} d_1 & d_1 z_1^\top \\ d_1 z_1 & S_1 + d_1 z_1 z_1^\top \end{pmatrix}\right) \\ d_1 \sim \text{IG}\left(\frac{\nu_0 + 2}{2}, \frac{\nu_0}{2}\right) \\ z_1 | S_1 \sim \mathcal{N}\left(\mu_1, \frac{S_1}{\nu_0}\right) \\ S_1 \sim \text{IW}(\nu_0 \bar{S}_1, \nu_0 + J + 1) \\ \beta \sim \mathcal{N}(0, \sigma_0^2) \end{cases} \quad (6)$$

Deriving $p(x | \gamma = 1)$. Write

$$x = \begin{pmatrix} x_1 \\ x_{-1} \end{pmatrix}.$$

Step 1: likelihood conditional on all latent variables. From the block normal distribution in (6),

$$p(x | \gamma = 1, \beta, d_1, z_1, S_1) = \mathcal{N}_J\left(x | \beta \begin{pmatrix} d_1 \\ d_1 z_1 \end{pmatrix}, \frac{1}{N} \begin{pmatrix} d_1 & d_1 z_1^\top \\ d_1 z_1 & S_1 + d_1 z_1 z_1^\top \end{pmatrix}\right).$$

Equivalently, using the conditional distribution of a block normal,

$$p(x | \gamma = 1, \beta, d_1, z_1, S_1) = \mathcal{N}\left(x_1 | \beta d_1, \frac{d_1}{N}\right) \mathcal{N}_{J-1}\left(x_{-1} | x_1 z_1, \frac{S_1}{N}\right).$$

The key simplification is that, **after conditioning on x_1 , the conditional mean of x_{-1} is $x_1 z_1$ and no longer contains β or d_1** (relevant to our earlier discussion).

Step 2: marginalize over $z_1 \mid S_1$. Since

$$z_1 \mid S_1 \sim \mathcal{N}_{J-1} \left(\mu_1, \frac{S_1}{\nu_0} \right)$$

and

$$x_{-1} \mid x_1, z_1, S_1 \sim \mathcal{N}_{J-1} \left(x_1 z_1, \frac{S_1}{N} \right),$$

we get

$$x_{-1} \mid x_1, S_1, \gamma = 1 \sim \mathcal{N}_{J-1} \left(x_1 \mu_1, \left(\frac{1}{N} + \frac{x_1^2}{\nu_0} \right) S_1 \right). \quad (7)$$

Therefore

$$p(x \mid \gamma = 1, \beta, d_1, S_1) = \mathcal{N} \left(x_1 \mid \beta d_1, \frac{d_1}{N} \right) \mathcal{N}_{J-1} \left(x_{-1} \mid x_1 \mu_1, \left(\frac{1}{N} + \frac{x_1^2}{\nu_0} \right) S_1 \right).$$

What I found so interesting is that the conditional distribution $p(x_{-1} \mid x_1, S_1, \gamma = 1)$ resembles what was suggested by Matthew last time (cf. (1)).

Step 3: marginalize over S_1 . Because $S_1 \sim \mathcal{IW}_{J-1}(\nu_0 \bar{S}_1, \nu_0 + J + 1)$, integrating S_1 gives a multivariate t distribution with $\nu_0 + 3$ degrees of freedom:

$$x_{-1} \mid x_1, \gamma = 1 \sim t_{J-1, \nu_0+3} \left(x_1 \mu_1, \frac{\nu_0}{\nu_0 + 3} \left(\frac{1}{N} + \frac{x_1^2}{\nu_0} \right) \bar{S}_1 \right),$$

where the second argument is the multivariate t scale matrix. Thus

$$p(x \mid \gamma = 1, \beta, d_1) = \mathcal{N} \left(x_1 \mid \beta d_1, \frac{d_1}{N} \right) t_{J-1, \nu_0+3} \left(x_{-1} \mid x_1 \mu_1, \frac{\nu_0}{\nu_0 + 3} \left(\frac{1}{N} + \frac{x_1^2}{\nu_0} \right) \bar{S}_1 \right).$$

Equivalently, the t factor is proportional to

$$|\bar{S}_1|^{-1/2} \left(\frac{1}{N} + \frac{x_1^2}{\nu_0} \right)^{-(J-1)/2} \left[1 + \frac{(x_{-1} - x_1 \mu_1)^\top (\nu_0 \bar{S}_1)^{-1} (x_{-1} - x_1 \mu_1)}{\frac{1}{N} + \frac{x_1^2}{\nu_0}} \right]^{-(\nu_0+J+2)/2}.$$

Using the block inverse formula (notice the block form of $\bar{R}$ (4)),

$$x^\top \bar{R}^{-1} x = x_1^2 + (x_{-1} - x_1 \mu_1)^\top \bar{S}_1^{-1} (x_{-1} - x_1 \mu_1),$$

so

$$1 + \frac{(x_{-1} - x_1 \mu_1)^\top (\nu_0 \bar{S}_1)^{-1} (x_{-1} - x_1 \mu_1)}{\frac{1}{N} + \frac{x_1^2}{\nu_0}} = \frac{x^\top \bar{R}^{-1} x + \nu_0/N}{x_1^2 + \nu_0/N}.$$

Therefore

$$T_1(x) \propto |\bar{S}_1|^{-1/2} \left(x_1^2 + \frac{\nu_0}{N} \right)^{(\nu_0+3)/2} \left(x^\top \bar{R}^{-1} x + \frac{\nu_0}{N} \right)^{-(\nu_0+J+2)/2}. \quad (8)$$

When comparing different values of γ , the final factor is common across SNPs. Also, since $\bar{R}_{11} = 1$, $|\bar{R}| = |\bar{S}_1|$; analogously, $|\bar{S}_j| = |\bar{R}|$ for any SNP j with $\bar{R}_{jj} = 1$.

Step 4: marginalize over β and d_1 . The preceding display separates the problem into an $x_{-1} \mid x_1$ factor and an x_1 factor. Define

$$T_1(x) = t_{J-1, \nu_0+3} \left(x_{-1} \mid x_1 \mu_1, \frac{\nu_0}{\nu_0+3} \left(\frac{1}{N} + \frac{x_1^2}{\nu_0} \right) \bar{S}_1 \right).$$

If we integrate over β , then

$$x_1 \mid d_1, \gamma = 1 \sim \mathcal{N} \left(0, \frac{d_1}{N} + d_1^2 \sigma_0^2 \right),$$

and hence (finally integrating over d_1):

$$p(x \mid \gamma = 1) = T_1(x) \int_0^\infty \mathcal{N} \left(x_1 \mid 0, \frac{d_1}{N} + d_1^2 \sigma_0^2 \right) \text{IG} \left(d_1; \frac{\nu_0+2}{2}, \frac{\nu_0}{2} \right) dd_1.$$

This is a one-dimensional integral over $d_1 > 0$. I do not see a standard closed form for it because the variance is $d_1/N + d_1^2 \sigma_0^2$. Replacing T_1 with its expression in (8),

$$\begin{aligned} p(x \mid \gamma = 1) &\propto |\bar{R}|^{-1/2} \left(x_1^2 + \frac{\nu_0}{N} \right)^{(\nu_0+3)/2} \left(x^\top \bar{R}^{-1} x + \frac{\nu_0}{N} \right)^{-(\nu_0+J+2)/2} \\ &\quad \times \int_0^\infty \mathcal{N} \left(x_1 \mid 0, \frac{d_1}{N} + d_1^2 \sigma_0^2 \right) d\text{IG}(d_1) \end{aligned}$$

Since $|\bar{R}|$ and $x^\top \bar{R}^{-1} x$ do not depend on which SNP is chosen, both factors cancel in the PIP normalization. Also,

$$\left(x_j^2 + \frac{\nu_0}{N} \right)^{(\nu_0+3)/2} = \left(\frac{\nu_0}{N} \right)^{(\nu_0+3)/2} \left(1 + \frac{N x_j^2}{\nu_0} \right)^{(\nu_0+3)/2},$$

and the first factor is common across SNPs. Hence,

$$p(\gamma = j \mid x) \propto \left(1 + \frac{N x_j^2}{\nu_0} \right)^{(\nu_0+3)/2} \int_0^\infty \mathcal{N} \left(x_j \mid 0, \frac{d_j}{N} + d_j^2 \sigma_0^2 \right) d\text{IG}(d_j).$$

Conclusion. In the original SER-love model, the marginal likelihood depends on $x^\top \bar{R}^{-1} x$, but this dependence is common across SNPs and cancels when computing the PIP. The resulting PIP weight depends on SNP j only through x_j .

2.2 Model B

Simplify $d_1 \equiv 1$ to avoid intractable integral (and we know the diagonal of the in-sample LD matrix is 1 anyway), we have the following model (model B):

$$\begin{cases} x \mid \beta, \gamma = 1, z_1, S_1 \sim \mathcal{N} \left(\beta \begin{pmatrix} 1 \\ z_1 \end{pmatrix}, \frac{1}{N} \begin{pmatrix} 1 & z_1^\top \\ z_1 & S_1 + z_1 z_1^\top \end{pmatrix} \right) \\ z_1 \mid S_1 \sim \mathcal{N} \left(\mu_1, \frac{S_1}{\nu_0} \right) \\ S_1 \sim \text{IW}(\nu_0 \bar{S}_1, \nu_0 + J + 1) \\ \beta \sim \mathcal{N}(0, \sigma_0^2) \end{cases} \quad (9)$$

This model leads to

$$p(x \mid \gamma = 1) \propto |\bar{R}|^{-1/2} \left(1 + \frac{Nx_j^2}{\nu_0}\right)^{(\nu_0+3)/2} \left(x^\top \bar{R}^{-1}x + \frac{\nu_0}{N}\right)^{-(\nu_0+J+2)/2} \mathcal{N}\left(x_1 \mid 0, \frac{1}{N} + \sigma_0^2\right).$$

Since $|\bar{R}|$ and $x^\top \bar{R}^{-1}x$ do not depend on which SNP is chosen, the PIP weights simplify further. Dropping the common factors gives

$$p(\gamma = j \mid x) \propto \left(1 + \frac{Nx_j^2}{\nu_0}\right)^{(\nu_0+3)/2} \mathcal{N}\left(x_j \mid 0, \frac{1}{N} + \sigma_0^2\right). \quad (10)$$

Compare to SER-RSS's PIP. Recall SER-RSS's PIP:

$$\tilde{p}(\gamma = j \mid x) \propto \exp\left(\frac{N\sigma_0^2}{N\sigma_0^2 + 1} \frac{Nx_j^2}{2}\right) \quad (11)$$

We now show that (10) tends to (11) as $\nu_0 \rightarrow \infty$. Since

$$\left(1 + \frac{Nx_j^2}{\nu_0}\right)^{(\nu_0+3)/2} = \left(1 + \frac{2}{\nu_0 + 3} \times \frac{\nu_0 + 3}{\nu_0} \frac{Nx_j^2}{2}\right)^{(\nu_0+3)/2} \asymp \exp\left\{\frac{\nu_0 + 3}{\nu_0} \frac{Nx_j^2}{2}\right\}$$

Also,

$$\mathcal{N}\left(x_j \mid 0, \frac{1}{N} + \sigma_0^2\right) \propto \exp\left\{-\frac{x_j^2}{2(1/N + \sigma_0^2)}\right\} = \exp\left\{-\frac{1}{1 + N\sigma_0^2} \frac{Nx_j^2}{2}\right\}.$$

Therefore the limiting weight in (10) is asymptotically (as ν_0 gets large)

$$\exp\left\{\left(\frac{\nu_0 + 3}{\nu_0} - \frac{1}{N\sigma_0^2 + 1}\right) \frac{Nx_j^2}{2}\right\} = \exp\left\{\left(\frac{N\sigma_0^2}{N\sigma_0^2 + 1} - \frac{1}{\nu_0 + 3}\right) \frac{Nx_j^2}{2}\right\} \approx \exp\left\{\frac{N\sigma_0^2}{N\sigma_0^2 + 1} \frac{Nx_j^2}{2}\right\}$$

which is exactly the SER-RSS PIP weight in (11). Hence, for finite ν_0 , PIP under SER-love model (B) is slightly more diffused than susie-rss.

Inference under model B (9). For a general SNP j , write

$$\mu_j = \bar{R}_{-j,j}, \quad \bar{S}_j = \bar{R}_{-j,-j} - \mu_j \mu_j^\top, \quad \kappa_j = \nu_0 + Nx_j^2.$$

Under model B, conditional on $\gamma = j$, the likelihood factorizes as

$$p(x \mid \beta, z_j, S_j, \gamma = j) = \mathcal{N}\left(x_j \mid \beta, \frac{1}{N}\right) \mathcal{N}_{J-1}\left(x_{-j} \mid x_j z_j, \frac{S_j}{N}\right).$$

Thus β is conditionally independent of (z_j, S_j) given x and $\gamma = j$, and

$$p(\beta, z_j, S_j \mid x, \gamma = j) = p(\beta \mid x_j, \gamma = j) p(z_j, S_j \mid x, \gamma = j).$$

- The posterior for β is the usual normal-normal update:

$$\beta \mid x, \gamma = j \sim \mathcal{N}\left(\frac{N}{N + \sigma_0^{-2}}x_j, \frac{1}{N + \sigma_0^{-2}}\right).$$

- For the LD-column variables, first condition on S_j . Combining

$$z_j \mid S_j \sim \mathcal{N}_{J-1}\left(\mu_j, \frac{S_j}{\nu_0}\right), \quad x_{-j} \mid x_j, z_j, S_j, \gamma = j \sim \mathcal{N}_{J-1}\left(x_j z_j, \frac{S_j}{N}\right)$$

gives

$$z_j \mid S_j, x, \gamma = j \sim \mathcal{N}_{J-1}\left(\frac{\nu_0 \mu_j + N x_j x_{-j}}{\nu_0 + N x_j^2}, \frac{S_j}{\nu_0 + N x_j^2}\right).$$

- Integrating out z_j gives

$$x_{-j} \mid x_j, S_j, \gamma = j \sim \mathcal{N}_{J-1}\left(x_j \mu_j, \left(\frac{1}{N} + \frac{x_j^2}{\nu_0}\right) S_j\right).$$

Therefore the posterior for S_j is inverse-Wishart:

$$S_j \mid x, \gamma = j \sim \mathcal{IW}\left(\nu_0 \bar{S}_j + \frac{N \nu_0}{\nu_0 + N x_j^2} (x_{-j} - x_j \mu_j)(x_{-j} - x_j \mu_j)^\top, \nu_0 + J + 2\right).$$

Equivalently, the joint posterior of (z_j, S_j) can be represented by first drawing S_j from this inverse-Wishart distribution and then drawing z_j from the conditional normal distribution above. Finally, the full posterior over γ uses the PIP weights in (10), normalized over $j = 1, \dots, J$.

Closed-form ELBO. Because the conditional posteriors above are exact, the optimized ELBO for model B is the exact marginal log likelihood. Let

$$V_\beta = \frac{1}{N} + \sigma_0^2, \quad q = x^\top \bar{R}^{-1} x.$$

For a uniform prior on γ , the optimized ELBO is

$$\mathcal{L}(\nu_0, \sigma_0^2) = \log \left\{ \frac{1}{J} \sum_{j=1}^J \exp(\ell_j) \right\},$$

where the component log marginal likelihood can be written as

$$\begin{aligned} \ell_j = & -\frac{1}{2} \log(2\pi V_\beta) - \frac{x_j^2}{2V_\beta} + \log \Gamma\left(\frac{\nu_0 + J + 2}{2}\right) - \log \Gamma\left(\frac{\nu_0 + 3}{2}\right) \\ & - \frac{J-1}{2} \log \pi - \frac{1}{2} \log |\bar{S}_j| + \frac{\nu_0 + 3}{2} \log\left(x_j^2 + \frac{\nu_0}{N}\right) - \frac{\nu_0 + J + 2}{2} \log\left(q + \frac{\nu_0}{N}\right). \end{aligned}$$

This is equivalent to the marginal likelihood derived above, but unlike the PIP formula it keeps the terms that are common across SNPs. Those common terms do not affect $\pi_j = p(\gamma = j \mid x)$ for fixed hyperparameters, but they do matter when learning ν_0 .

Empirical Bayes learning of ν_0 and σ_0^2 . The empirical Bayes estimates can be obtained by maximizing the closed-form objective

$$(\hat{\nu}_0, \hat{\sigma}_0^2) = \arg \max_{\nu_0 > 0, \sigma_0^2 \geq 0} \mathcal{L}(\nu_0, \sigma_0^2).$$

After choosing $(\hat{\nu}_0, \hat{\sigma}_0^2)$, the posterior inclusion probabilities are

$$\pi_j = \frac{\exp(\ell_j)}{\sum_{k=1}^J \exp(\ell_k)} \Big|_{\nu_0 = \hat{\nu}_0, \sigma_0^2 = \hat{\sigma}_0^2}.$$

Equivalently, for computing PIPs at fixed hyperparameters one may use the simplified weights in (10); for empirical Bayes optimization one should use $\mathcal{L}(\nu_0, \sigma_0^2)$ above, because it retains the ν_0 -dependent normalizing constants and the common $\log(q + \nu_0/N)$ term.

EM view of empirical Bayes. Another way to use the fact that the posterior is exact is to write the usual ELBO identity. Let

$$\theta = (\gamma, \beta, z_\gamma, S_\gamma), \quad q(\theta) = p(\theta \mid x; \nu_0^{\text{old}}, \sigma_{0,\text{old}}^2).$$

Strictly speaking,

$$\log p(x; \nu_0, \sigma_0^2) \geq \mathbb{E}_q\{\log p(x \mid \theta)\} - \text{KL}\{q(\theta) \parallel p(\theta; \nu_0, \sigma_0^2)\},$$

with equality when q is the posterior under the same hyperparameters. In an EM algorithm, the E-step fixes q at the current hyperparameters and the M-step maximizes

$$Q(\nu_0, \sigma_0^2) = \mathbb{E}_q\{\log p(\theta; \nu_0, \sigma_0^2)\},$$

because the likelihood term does not involve ν_0 or σ_0^2 in model B.

This gives a closed-form update for σ_0^2 . If

$$\pi_j = q(\gamma = j), \quad m_{\beta j} = \frac{N}{N + \sigma_{0,\text{old}}^{-2}} x_j, \quad v_\beta = \frac{1}{N + \sigma_{0,\text{old}}^{-2}},$$

then

$$\sigma_{0,\text{new}}^2 = \sum_{j=1}^J \pi_j (m_{\beta j}^2 + v_\beta).$$

For ν_0 , the same EM view reduces the update to a one-dimensional optimization, but not to a closed-form expression. Let $p = J - 1$ and write the current posterior of S_j as

$$S_j \mid x, \gamma = j \sim \mathcal{IW}_p(\Psi_j, m_j), \quad m_j = \nu_0^{\text{old}} + J + 2,$$

where

$$\Psi_j = \nu_0^{\text{old}} \bar{S}_j + \frac{N \nu_0^{\text{old}}}{\nu_0^{\text{old}} + N x_j^2} (x_{-j} - x_j \mu_j)(x_{-j} - x_j \mu_j)^\top.$$

Also let

$$\bar{z}_j = \frac{\nu_0^{\text{old}} \mu_j + N x_j x_{-j}}{\nu_0^{\text{old}} + N x_j^2}, \quad \kappa_j^{\text{old}} = \nu_0^{\text{old}} + N x_j^2.$$

These quantities can be evaluated without forming or inverting the $(J-1) \times (J-1)$ matrices Ψ_j . Let

$$\nu_\star = \nu_0^{\text{old}}, \quad q_R = x^\top \bar{R}^{-1} x, \quad \delta_j = x_{-j} - x_j \mu_j.$$

Then

$$\Psi_j = \nu_\star \bar{S}_j + \frac{N \nu_\star}{\kappa_j^{\text{old}}} \delta_j \delta_j^\top.$$

By the Woodbury identity,

$$\Psi_j^{-1} = \frac{1}{\nu_\star} \bar{S}_j^{-1} - \frac{N}{\nu_\star(\nu_\star + N q_R)} \bar{S}_j^{-1} \delta_j \delta_j^\top \bar{S}_j^{-1},$$

because

$$1 + \frac{N}{\kappa_j^{\text{old}}} (q_R - x_j^2) = \frac{\nu_\star + N q_R}{\kappa_j^{\text{old}}}.$$

The determinant lemma gives

$$\log |\Psi_j| = (J-1) \log \nu_\star + \log |\bar{S}_j| + \log \left(\frac{\nu_\star + N q_R}{\kappa_j^{\text{old}}} \right).$$

Since $\bar{R}_{jj} = 1$, $\log |\bar{S}_j| = \log |\bar{R}|$, so this is also common to compute once.

The needed posterior expectations are

$$A_j = \mathbb{E}_q\{\text{tr}(\bar{S}_j S_j^{-1})\} = m_j \left\{ \frac{J-1}{\nu_\star} - \frac{N(q_R - x_j^2)}{\nu_\star(\nu_\star + N q_R)} \right\},$$

$$B_j = \mathbb{E}_q\{(z_j - \mu_j)^\top S_j^{-1} (z_j - \mu_j)\} = m_j \frac{N^2 x_j^2 (q_R - x_j^2)}{\nu_\star \kappa_j^{\text{old}} (\nu_\star + N q_R)} + \frac{p}{\kappa_j^{\text{old}}},$$

and

$$C_j = \mathbb{E}_q\{\log |S_j|\} = (J-1) \log \nu_\star + \log |\bar{R}| + \log \left(\frac{\nu_\star + N q_R}{\kappa_j^{\text{old}}} \right) - \sum_{r=1}^p \psi \left(\frac{m_j + 1 - r}{2} \right) - p \log 2.$$

Thus all SNP-specific parts of A_j, B_j, C_j depend on j only through x_j^2 and $q_R - x_j^2$. In implementation, one can compute q_R and $\log |\bar{R}|$ once, then evaluate the remaining terms with scalar operations for each SNP. Up to constants that do not depend on ν_0 , the EM objective for ν_0 is

$$\begin{aligned} Q_\nu(\nu_0) = \sum_{j=1}^J \pi_j \left[\frac{p}{2} \log \nu_0 + \frac{\nu_0 + J + 1}{2} \log |\nu_0 \bar{S}_j| - \frac{(\nu_0 + J + 1)p}{2} \log 2 \right. \\ \left. - \log \Gamma_p \left(\frac{\nu_0 + J + 1}{2} \right) - \frac{\nu_0 + 2J + 2}{2} C_j - \frac{\nu_0}{2} (A_j + B_j) \right]. \end{aligned}$$

Thus

$$\boxed{\nu_{0,\text{new}} = \arg \max_{\nu_0 > 0} Q_\nu(\nu_0).}$$

So the exact-posterior ELBO gives a particularly nice update for σ_0^2 , and a stable scalar optimization for ν_0 . For ν_0 , the closed-form marginal likelihood $\mathcal{L}(\nu_0, \sigma_0^2)$ above is often even simpler to optimize directly, but the EM form is useful for deriving a monotone coordinate-ascent algorithm.

2.3 Other representations of Model B

Recall Model B:

$$\begin{cases} x|\beta, \gamma = 1, z_1, S_1 \sim \mathcal{N}\left(\beta \begin{pmatrix} 1 \\ z_1 \end{pmatrix}, \frac{1}{N} \begin{pmatrix} 1 & z_1^\top \\ z_1 & S_1 + z_1 z_1^\top \end{pmatrix}\right) \\ z_1|S_1 \sim \mathcal{N}\left(\mu_1, \frac{S_1}{\nu_0}\right) \\ S_1 \sim \mathcal{IW}(\nu_0 \bar{S}_1, \nu_0 + J + 1) \\ \beta \sim \mathcal{N}(0, \sigma_0^2) \end{cases} \quad (12)$$

or equivalently,

$$\begin{cases} x_1 | \beta, \gamma = 1 \sim \mathcal{N}\left(\beta, \frac{1}{N}\right), \\ x_{-1} | x_1, z_1, \gamma = 1 \sim \mathcal{N}_{J-1}\left(x_1 z_1, \frac{1}{N} S_1\right), \\ z_1|S_1 \sim \mathcal{N}\left(\mu_1, \frac{S_1}{\nu_0}\right), \\ S_1 \sim \mathcal{IW}(\nu_0 \bar{S}_1, \nu_0 + J + 1), \\ \beta \sim \mathcal{N}(0, \sigma_0^2). \end{cases} \quad (13)$$

Here are three other equivalent representations of model B for $\gamma = 1$. Write

$$\mu_1 = \bar{R}_{-1,1}, \quad \bar{S}_1 = \bar{R}_{-1,-1} - \mu_1 \mu_1^\top.$$

1. After integrating out $z_1 | S_1$.

$$\begin{cases} x_1 | \beta, \gamma = 1 \sim \mathcal{N}\left(\beta, \frac{1}{N}\right), \\ x_{-1} | x_1, S_1, \gamma = 1 \sim \mathcal{N}_{J-1}\left(x_1 \mu_1, \left(\frac{1}{N} + \frac{x_1^2}{\nu_0}\right) S_1\right), \\ S_1 \sim \mathcal{IW}(\nu_0 \bar{S}_1, \nu_0 + J + 1), \\ \beta \sim \mathcal{N}(0, \sigma_0^2). \end{cases} \quad (14)$$

2. After further integrating out S_1 .

$$\begin{cases} x_1 | \beta, \gamma = 1 \sim \mathcal{N}\left(\beta, \frac{1}{N}\right), \\ x_{-1} | x_1, \gamma = 1 \sim t_{J-1, \nu_0+3}\left(x_1 \mu_1, \frac{\nu_0}{\nu_0+3} \left(\frac{1}{N} + \frac{x_1^2}{\nu_0}\right) \bar{S}_1\right), \\ \beta \sim \mathcal{N}(0, \sigma_0^2). \end{cases} \quad (15)$$

3. Using the normal–Gamma mixture for the t distribution. With shape–rate parameterization, Conditional on ω_1 :

$$\begin{cases} x_1 \mid \beta, \gamma = 1 \sim \mathcal{N}\left(\beta, \frac{1}{N}\right), \\ x_{-1} \mid x_1, \omega_1, \gamma = 1 \sim \mathcal{N}_{J-1}\left(x_1\mu_1, \frac{1}{\omega_1} \left(\frac{1}{N} + \frac{x_1^2}{\nu_0}\right) \bar{S}_1\right), \\ \omega_1 \sim \text{Gamma}\left(\frac{\nu_0 + 3}{2}, \frac{\nu_0}{2}\right), \\ \beta \sim \mathcal{N}(0, \sigma_0^2). \end{cases} \quad (16)$$

2.4 Try to achieve common noise

I am thinking about another representation of model B (9) that will be useful in extending it to a Susie-like model. In particular, it would be useful to have a model with a common noise among all choices of $\gamma = j$.

An exact two-error view of model B. For $\gamma = 1$, model B can be viewed as a SER model with two additional errors, but the variance error is not an independent additive Gaussian error. Let $A_1 = A_1(S_1)$ be any matrix such that

$$A_1 \bar{S}_1 A_1^\top = S_1.$$

Let

$$e_1 \sim \mathcal{N}\left(0, \frac{1}{N}\right), \quad e_{-1} \sim \mathcal{N}_{J-1}\left(0, \frac{\bar{S}_1}{N}\right), \quad h_1 \sim \mathcal{N}_{J-1}\left(0, \frac{\bar{S}_1}{\nu_0}\right),$$

mutually independent conditional on S_1 , and set

$$\begin{cases} x_1 = \beta + e_1, \\ x_{-1} = x_1\mu_1 + A_1 e_{-1} + x_1 A_1 h_1, \\ S_1 \sim \mathcal{IW}(\nu_0 \bar{S}_1, \nu_0 + J + 1), \\ \beta \sim \mathcal{N}(0, \sigma_0^2). \end{cases} \quad (17)$$

The term $x_1 A_1 h_1$ is the error in the mean/LD column, because $A_1 h_1 \mid S_1 \sim \mathcal{N}_{J-1}(0, S_1/\nu_0)$. The term $A_1 e_{-1}$ is the variance error: it transforms the SER conditional noise $e_{-1} \sim \mathcal{N}_{J-1}(0, \bar{S}_1/N)$ into

$$A_1 e_{-1} \mid S_1 \sim \mathcal{N}_{J-1}\left(0, \frac{S_1}{N}\right).$$

Consequently, after integrating out e_{-1} and h_1 ,

$$x_{-1} \mid x_1, S_1, \gamma = 1 \sim \mathcal{N}_{J-1}\left(x_1\mu_1, \left(\frac{1}{N} + \frac{x_1^2}{\nu_0}\right) S_1\right),$$

which is exactly (14) in block form.

Equivalently, if $x^{\text{SER}} \mid \beta, \gamma = 1 \sim \mathcal{N}_J(\beta \bar{R}_{\cdot 1}, \bar{R}/N)$, then the conditional SER residual in the non-causal coordinates is $e_{-1} = x_{-1}^{\text{SER}} - x_1\mu_1$, and

$$x = x^{\text{SER}} + u_1 \{(A_1 - I)e_{-1} + x_1 A_1 h_1\}.$$

This is the desired “SER plus two errors” picture. The caveat is that $(A_1 - I)e_{-1}$ is coupled to the SER noise; it is not an independent additive Gaussian error. This coupling is what avoids the impossible requirement that $S_1 - \bar{S}_1$ be positive semidefinite.

What is the distribution of A_1 ? It is not unique unless we choose a particular square-root convention. If

$$T_1 = \bar{S}_1^{-1/2} S_1 \bar{S}_1^{-1/2},$$

then by affine equivariance of the inverse-Wishart distribution,

$$T_1 \sim \mathcal{IW}_{J-1}(\nu_0 I, \nu_0 + J + 1).$$

Every valid A_1 can be written as

$$A_1 = \bar{S}_1^{1/2} B_1 \bar{S}_1^{-1/2}, \quad B_1 B_1^\top = T_1.$$

Thus one canonical choice is

$$B_1 = T_1^{1/2}, \quad A_1 = \bar{S}_1^{1/2} T_1^{1/2} \bar{S}_1^{-1/2}.$$

Under this convention, A_1 has the distribution induced by the symmetric matrix square root of an isotropic inverse-Wishart random matrix. This does not have a simpler standard named form, but it is concentrated around I as ν_0 grows. Other choices, such as $B_1 = T_1^{1/2} O_1$ for an orthogonal matrix O_1 , give the same S_1 but a different distribution for A_1 .

Approximating the variance of x_1 . Starting from (16), try the approximation

$$x_1 \mid \beta, \omega_1, \gamma = 1 \sim \mathcal{N}\left(\beta, \frac{1}{N\omega_1}\right),$$

while keeping the conditional model for x_{-1} given x_1 and ω_1 :

$$\begin{cases} x_1 \mid \beta, \omega_1, \gamma = 1 \sim \mathcal{N}\left(\beta, \frac{1}{N\omega_1}\right), \\ x_{-1} \mid x_1, \omega_1, \gamma = 1 \sim \mathcal{N}_{J-1}\left(x_1 \mu_1, \frac{1}{\omega_1} \left(\frac{1}{N} + \frac{x_1^2}{\nu_0}\right) \bar{S}_1\right), \\ \omega_1 \sim \text{Gamma}\left(\frac{\nu_0 + 3}{2}, \frac{\nu_0}{2}\right), \\ \beta \sim \mathcal{N}(0, \sigma_0^2). \end{cases} \quad (18)$$

This approximation has a clean whole-vector representation. Let

$$\zeta_1 \mid x_1, \omega_1, \gamma = 1 \sim \mathcal{N}_{J-1}\left(0, \frac{x_1^2}{\nu_0 \omega_1} \bar{S}_1\right).$$

Then (18) is equivalent to

$$\begin{cases} x \mid \beta, \zeta_1, \omega_1, \gamma = 1 \sim \mathcal{N}_J\left(\beta \bar{R}_{\cdot 1} + u_1(\zeta_1), \frac{\bar{R}}{N\omega_1}\right), \\ \zeta_1 \mid x_1, \omega_1, \gamma = 1 \sim \mathcal{N}_{J-1}\left(0, \frac{x_1^2}{\nu_0 \omega_1} \bar{S}_1\right), \\ \omega_1 \sim \text{Gamma}\left(\frac{\nu_0 + 3}{2}, \frac{\nu_0}{2}\right), \\ \beta \sim \mathcal{N}(0, \sigma_0^2). \end{cases} \quad (19)$$

Indeed, the conditional distribution induced by the common noise $\bar{R}/(N\omega_1)$ is

$$x_1 \mid \beta, \omega_1, \gamma = 1 \sim \mathcal{N}\left(\beta, \frac{1}{N\omega_1}\right), \quad x_{-1} \mid x_1, \zeta_1, \omega_1, \gamma = 1 \sim \mathcal{N}_{J-1}\left(x_1\mu_1 + \zeta_1, \frac{\bar{S}_1}{N\omega_1}\right).$$

Integrating out ζ_1 gives the second line of (18).

Thus this approximation achieves a common-noise representation with common noise $\bar{R}/(N\omega_1)$. This is not the same as fixed noise $\bar{R}/N$. However, if we are willing to use the random common-noise scale $\bar{R}/(N\omega_1)$, then (19) is a valid representation.

PIP under the random common-noise model. For a general candidate SNP j , define

$$\mu_j = \bar{R}_{-j,j}, \quad \bar{S}_j = \bar{R}_{-j,-j} - \mu_j \mu_j^\top, \quad c_j = \frac{1}{N} + \frac{x_j^2}{\nu_0}, \quad q_R = x^\top \bar{R}^{-1} x.$$

The block inverse formula gives

$$(x_{-j} - x_j \mu_j)^\top \bar{S}_j^{-1} (x_{-j} - x_j \mu_j) = q_R - x_j^2.$$

Under (18), after integrating out β but conditioning on ω_j ,

$$x_j \mid \omega_j, \gamma = j \sim \mathcal{N}\left(0, \sigma_0^2 + \frac{1}{N\omega_j}\right),$$

and

$$x_{-j} \mid x_j, \omega_j, \gamma = j \sim \mathcal{N}_{J-1}\left(x_j \mu_j, \frac{c_j}{\omega_j} \bar{S}_j\right).$$

Therefore

$$\begin{aligned} p(x \mid \gamma = j) &= \int_0^\infty \mathcal{N}\left(x_j; 0, \sigma_0^2 + \frac{1}{N\omega}\right) \mathcal{N}_{J-1}\left(x_{-j}; x_j \mu_j, \frac{c_j}{\omega} \bar{S}_j\right) \\ &\quad \times \frac{(\nu_0/2)^{(\nu_0+3)/2}}{\Gamma((\nu_0+3)/2)} \omega^{(\nu_0+3)/2-1} \exp\left(-\frac{\nu_0\omega}{2}\right) d\omega \\ &= (2\pi)^{-J/2} |\bar{S}_j|^{-1/2} c_j^{-(J-1)/2} \frac{(\nu_0/2)^{(\nu_0+3)/2}}{\Gamma((\nu_0+3)/2)} \\ &\quad \times \int_0^\infty \left(\sigma_0^2 + \frac{1}{N\omega}\right)^{-1/2} \omega^{(\nu_0+3)/2+(J-1)/2-1} \\ &\quad \times \exp\left\{-\frac{x_j^2}{2(\sigma_0^2 + 1/(N\omega))} - \frac{\omega(q_R - x_j^2)}{2c_j} - \frac{\nu_0\omega}{2}\right\} d\omega. \end{aligned} \quad (20)$$

Since $\bar{R}_{jj} = 1$, $|\bar{S}_j| = |\bar{R}|$, so the determinant factor is common across SNPs. Dropping constants common in j , the PIP weights are

$$\begin{aligned} w_j^{\text{scaled}} &= c_j^{-(J-1)/2} \int_0^\infty \left(\sigma_0^2 + \frac{1}{N\omega}\right)^{-1/2} \omega^{(\nu_0+J)/2} \\ &\quad \times \exp\left\{-\frac{x_j^2}{2(\sigma_0^2 + 1/(N\omega))} - \frac{\omega(q_R - x_j^2)}{2c_j} - \frac{\nu_0\omega}{2}\right\} d\omega, \quad \pi_j = \frac{w_j^{\text{scaled}}}{\sum_{k=1}^J w_k^{\text{scaled}}}. \end{aligned} \quad (21)$$

Thus the random common-noise model is computationally a one-dimensional integral per SNP, but it does not retain the exact irrelevance property. The common statistic $q_R = x^\top \bar{R}^{-1} x$ appears inside the integral with the SNP-specific coefficient $1/c_j$, so it cannot be factored out and cancelled in the PIP normalization.

A variant that scales the mean of x_1 . Another possibility, closer in spirit to model A, is to replace the first line of (16) by

$$x_1 \mid \beta, \omega_1, \gamma = 1 \sim \mathcal{N}\left(\frac{\beta}{\sqrt{\omega_1}}, \frac{1}{N\omega_1}\right).$$

Keeping the same conditional distribution of x_{-1} , the block model becomes

$$\begin{cases} x_1 \mid \beta, \omega_1, \gamma = 1 \sim \mathcal{N}\left(\frac{\beta}{\sqrt{\omega_1}}, \frac{1}{N\omega_1}\right), \\ x_{-1} \mid x_1, \omega_1, \gamma = 1 \sim \mathcal{N}_{J-1}\left(x_1\mu_1, \frac{1}{\omega_1}\left(\frac{1}{N} + \frac{x_1^2}{\nu_0}\right)\bar{S}_1\right), \\ \omega_1 \sim \text{Gamma}\left(\frac{\nu_0 + 3}{2}, \frac{\nu_0}{2}\right), \\ \beta \sim \mathcal{N}(0, \sigma_0^2). \end{cases} \quad (22)$$

This has the common-noise representation

$$\begin{cases} x \mid \beta, \zeta_1, \omega_1, \gamma = 1 \sim \mathcal{N}_J\left(\frac{\beta}{\sqrt{\omega_1}}\bar{R}_{\cdot 1} + u_1(\zeta_1), \frac{\bar{R}}{N\omega_1}\right), \\ \zeta_1 \mid x_1, \omega_1, \gamma = 1 \sim \mathcal{N}_{J-1}\left(0, \frac{x_1^2}{\nu_0\omega_1}\bar{S}_1\right), \\ \omega_1 \sim \text{Gamma}\left(\frac{\nu_0 + 3}{2}, \frac{\nu_0}{2}\right), \\ \beta \sim \mathcal{N}(0, \sigma_0^2). \end{cases} \quad (23)$$

The only difference from (19) is that the effect mean is also divided by $\sqrt{\omega_1}$. If we define $\alpha_1 = \beta/\sqrt{\omega_1}$, then

$$\alpha_1 \mid \omega_1 \sim \mathcal{N}\left(0, \frac{\sigma_0^2}{\omega_1}\right),$$

so the prior variance of the effective effect size has the same $1/\omega_1$ scaling as the common noise. This is the useful feature of the $1/\sqrt{\omega_1}$ scaling.

The PIP calculation is almost the same as above. Conditional on ω_j and after integrating out β ,

$$x_j \mid \omega_j, \gamma = j \sim \mathcal{N}\left(0, \frac{\sigma_0^2 + 1/N}{\omega_j}\right),$$

whereas

$$x_{-j} \mid x_j, \omega_j, \gamma = j \sim \mathcal{N}_{J-1}\left(x_j\mu_j, \frac{c_j}{\omega_j}\bar{S}_j\right)$$

is unchanged. Let

$$V_\beta = \sigma_0^2 + \frac{1}{N}, \quad a = \frac{\nu_0 + 3}{2}, \quad b = \frac{\nu_0}{2}.$$

Then

$$\begin{aligned}
p(x \mid \gamma = j) &= (2\pi)^{-J/2} |\bar{S}_j|^{-1/2} V_\beta^{-1/2} c_j^{-(J-1)/2} \frac{b^a}{\Gamma(a)} \\
&\quad \times \int_0^\infty \omega^{a+J/2-1} \exp \left[-\omega \left\{ b + \frac{x_j^2}{2V_\beta} + \frac{q_R - x_j^2}{2c_j} \right\} \right] d\omega \\
&= (2\pi)^{-J/2} |\bar{S}_j|^{-1/2} V_\beta^{-1/2} c_j^{-(J-1)/2} \frac{b^a \Gamma(a + J/2)}{\Gamma(a)} \\
&\quad \times \left\{ b + \frac{x_j^2}{2V_\beta} + \frac{q_R - x_j^2}{2c_j} \right\}^{-(a+J/2)}.
\end{aligned} \tag{24}$$

Since $|\bar{S}_j| = |\bar{R}|$ is common across SNPs, the PIP weights are

$$\begin{aligned}
w_j^{\text{sqrt-scaled}} &= c_j^{-(J-1)/2} \left\{ \frac{\nu_0}{2} + \frac{x_j^2}{2(\sigma_0^2 + 1/N)} + \frac{q_R - x_j^2}{2c_j} \right\}^{-(\nu_0 + J + 3)/2}, \\
\pi_j &= \frac{w_j^{\text{sqrt-scaled}}}{\sum_{k=1}^J w_k^{\text{sqrt-scaled}}}.
\end{aligned} \tag{25}$$

This is better computationally: the integral over ω_j is now conjugate and has a closed form. It is also closer to model A in the sense that both the effective signal and the common noise share the same $1/\omega_j$ scale after marginalizing over β . However, it still does not recover the exact null-SNP irrelevance property, because the common statistic q_R appears with the SNP-specific coefficient $1/c_j$ inside the final bracket.

3 Extension to susie

Now we try to extend model B into a Susie-like model, where we allow more than one non-zero effect. A natural extension is to keep the usual SuSiE additive decomposition, but replace each single-effect update by the model B single-effect calculation above.

3.1 A first proposed multi-effect model

Let there be L single-effect components. For each component ℓ , introduce a causal index γ_ℓ , an effect size β_ℓ , and a relaxed LD column $z_{\ell j}$ when $\gamma_\ell = j$. A direct extension of model B is

$$x \mid \{\gamma_\ell, \beta_\ell, z_{\ell, \gamma_\ell}, S_{\ell, \gamma_\ell}\}_{\ell=1}^L \sim \mathcal{N}_J \left(\sum_{\ell=1}^L \beta_\ell u_{\gamma_\ell}(z_{\ell, \gamma_\ell}), \frac{\bar{R}}{N} \right),$$

where $u_j(z_j)$ is the vector with entry 1 at coordinate j and entries z_j on the remaining coordinates. The priors are

$$\gamma_\ell \sim \text{Unif}\{1, \dots, J\}, \quad \beta_\ell \sim \mathcal{N}(0, \sigma_0^2),$$

and, conditional on $\gamma_\ell = j$,

$$z_{\ell j} \mid S_{\ell j} \sim \mathcal{N}_{J-1} \left(\mu_j, \frac{S_{\ell j}}{\nu_0} \right), \quad S_{\ell j} \sim \text{IW}_{J-1}(\nu_0 \bar{S}_j, \nu_0 + J + 1).$$

This model gives each single-effect component its own relaxed active LD column. This is the closest analogue of SuSiE: multiple effects add in the mean, while each effect has a one-hot index and a scalar coefficient.

Sanity check: the $L = 1$ case. This first proposed model is actually not the right extension of model B. The issue is already visible when $L = 1$. Take $\gamma = 1$. Under the proposed model,

$$x \mid \beta, z_1, S_1, \gamma = 1 \sim \mathcal{N}_J \left(\beta \begin{pmatrix} 1 \\ z_1 \end{pmatrix}, \frac{\bar{R}}{N} \right),$$

so the covariance is fixed at $\bar{R}/N$ rather than using the (z_1, S_1) -dependent covariance in model B. Using the block form

$$\bar{R} = \begin{pmatrix} 1 & \mu_1^\top \\ \mu_1 & \bar{S}_1 + \mu_1 \mu_1^\top \end{pmatrix},$$

we get

$$x_1 \mid \beta, \gamma = 1 \sim \mathcal{N} \left(\beta, \frac{1}{N} \right),$$

and

$$x_{-1} \mid x_1, \beta, z_1, \gamma = 1 \sim \mathcal{N}_{J-1} \left(x_1 \mu_1 + \beta(z_1 - \mu_1), \frac{\bar{S}_1}{N} \right).$$

This is already different from model B, where the conditional mean is $x_1 z_1$ and does not contain β .

Now integrate over $z_1 \mid S_1$. Since

$$z_1 \mid S_1 \sim \mathcal{N}_{J-1} \left(\mu_1, \frac{S_1}{\nu_0} \right),$$

we obtain

$$x_{-1} \mid x_1, \beta, S_1, \gamma = 1 \sim \mathcal{N}_{J-1} \left(x_1 \mu_1, \frac{\bar{S}_1}{N} + \frac{\beta^2}{\nu_0} S_1 \right).$$

Therefore

$$\begin{aligned} p(x \mid \gamma = 1) &= \mathcal{N} \left(x_1 \mid 0, \frac{1}{N} + \sigma_0^2 \right) \\ &\quad \times \mathbb{E}_{\beta \mid x_1} \mathbb{E}_{S_1} \left[\mathcal{N}_{J-1} \left(x_{-1} \mid x_1 \mu_1, \frac{\bar{S}_1}{N} + \frac{\beta^2}{\nu_0} S_1 \right) \right], \end{aligned}$$

where

$$\beta \mid x_1 \sim \mathcal{N} \left(\frac{N}{N + \sigma_0^{-2}} x_1, \frac{1}{N + \sigma_0^{-2}} \right), \quad S_1 \sim \mathcal{IW}_{J-1}(\nu_0 \bar{S}_1, \nu_0 + J + 1).$$

Thus the PIP weight under this proposed model is

$$\boxed{p(\gamma = j \mid x) \propto \mathcal{N} \left(x_j \mid 0, \frac{1}{N} + \sigma_0^2 \right) \mathbb{E}_{\beta \mid x_j} \mathbb{E}_{S_j} \left[\mathcal{N}_{J-1} \left(x_{-j} \mid x_j \mu_j, \frac{\bar{S}_j}{N} + \frac{\beta^2}{\nu_0} S_j \right) \right]}.$$

This is not the model B PIP. In particular, it depends on x_{-j} through the conditional residual $x_{-j} - x_j \mu_j$, and the integral is not conjugate because the covariance is a sum of the fixed matrix $\bar{S}_j/N$ and the random matrix $(\beta^2/\nu_0)S_j$. Therefore this first proposed multi-effect generative model does not preserve the single-effect irrelevance property we want.

3.2 Variational approximation

The failed $L = 1$ check above means that we should not simply use the fixed noise covariance $\bar{R}/N$ and then plug in the model B single-effect weights. Still, it is useful to record what a SuSiE-like variational strategy would have to look like. As in SuSiE, the exact posterior is not tractable when $L > 1$ because the effects interact through the shared likelihood. A natural mean-field approximation is

$$q(\{\gamma_\ell, \beta_\ell, z_{\ell, \gamma_\ell}, S_{\ell, \gamma_\ell}\}_{\ell=1}^L) = \prod_{\ell=1}^L q_\ell(\gamma_\ell, \beta_\ell, z_{\ell, \gamma_\ell}, S_{\ell, \gamma_\ell}).$$

Each factor q_ℓ can be updated by holding the other components fixed in expectation. Define the expected contribution of component k by

$$\hat{b}_k = \mathbb{E}_{q_k}\{\beta_k u_{\gamma_k}(z_{k, \gamma_k})\},$$

and define the residual for updating component ℓ as

$$x^{(\ell)} = x - \sum_{k \neq \ell} \hat{b}_k.$$

The tempting coordinate update would be to apply the single-effect model B formulas to $x^{(\ell)}$ in place of x . The sanity check above shows that this update is not justified by the first proposed generative model with fixed noise covariance $\bar{R}/N$. Instead, it should be viewed as a possible pseudo-likelihood or variational working update that deliberately preserves the model B single-effect behavior.

In particular, for each candidate SNP j , let $x_j^{(\ell)}$ be the j th entry of the residual and let

$$q_\ell^R = x^{(\ell)\top} \bar{R}^{-1} x^{(\ell)}.$$

The single-effect weight for component ℓ is

$$w_{\ell j} \propto \left(1 + \frac{N x_j^{(\ell)2}}{\nu_0}\right)^{(\nu_0+3)/2} \mathcal{N}\left(x_j^{(\ell)} \mid 0, \frac{1}{N} + \sigma_0^2\right),$$

and the variational PIP for component ℓ is

$$\alpha_{\ell j} = \frac{w_{\ell j}}{\sum_{k=1}^J w_{\ell k}}.$$

Conditional on $\gamma_\ell = j$, the variational posteriors for $\beta_\ell, z_{\ell j}, S_{\ell j}$ are exactly the model B posterior formulas with x replaced by $x^{(\ell)}$.

3.3 Expected fitted effect

For the SuSiE-style coordinate ascent algorithm, we also need the expected fitted vector $\hat{b}_\ell$. Conditional on $\gamma_\ell = j$,

$$\mathbb{E}(\beta_\ell \mid x^{(\ell)}, \gamma_\ell = j) = \frac{N}{N + \sigma_0^{-2}} x_j^{(\ell)}.$$

Also,

$$\mathbb{E}(z_{\ell j} \mid x^{(\ell)}, \gamma_\ell = j) = \frac{\nu_0 \mu_j + N x_j^{(\ell)} x_{-j}^{(\ell)}}{\nu_0 + N x_j^{(\ell)2}}.$$

Because β_ℓ and $(z_{\ell j}, S_{\ell j})$ factorize conditional on $x^{(\ell)}$ and $\gamma_\ell = j$ in the single-effect update, we can use

$$\mathbb{E}\{\beta_\ell u_j(z_{\ell j}) \mid x^{(\ell)}, \gamma_\ell = j\} = m_{\beta, \ell j} \begin{pmatrix} 1 \\ m_{z, \ell j} \end{pmatrix}_j,$$

where

$$m_{\beta, \ell j} = \frac{N}{N + \sigma_0^{-2}} x_j^{(\ell)}, \quad m_{z, \ell j} = \frac{\nu_0 \mu_j + N x_j^{(\ell)} x_{-j}^{(\ell)}}{\nu_0 + N x_j^{(\ell)2}}.$$

Here the subscript j on the vector means that the scalar 1 is placed in coordinate j and $m_{z, \ell j}$ fills the remaining coordinates. Therefore

$$\hat{b}_\ell = \sum_{j=1}^J \alpha_{\ell j} m_{\beta, \ell j} \begin{pmatrix} 1 \\ m_{z, \ell j} \end{pmatrix}_j.$$

3.4 What is exact and what is approximate

For $L = 1$, this reduces exactly to model B. For $L > 1$, the coordinate update is a SuSiE-style variational approximation: each update treats the current residual as the data for one single-effect model. This is not the exact posterior under the full multi-effect model, but it preserves the key single-effect behavior. In particular, the candidate weight $w_{\ell j}$ depends on the residual only through the marginal residual statistic $x_j^{(\ell)}$, so the update retains the desired null-SNP irrelevance property at the level of each single-effect regression.

3.5 Hyperparameter learning

The empirical Bayes updates for σ_0^2 and ν_0 can also be lifted from the single-effect case by averaging over components. For example, the EM-style update for σ_0^2 becomes

$$\sigma_{0, \text{new}}^2 = \frac{1}{L} \sum_{\ell=1}^L \sum_{j=1}^J \alpha_{\ell j} (m_{\beta, \ell j}^2 + v_\beta), \quad v_\beta = \frac{1}{N + \sigma_{0, \text{old}}^{-2}}.$$

For ν_0 , one can use the same scalar EM objective as in the single-effect case, replacing the sum over j by a weighted sum over (ℓ, j) and using the residual $x^{(\ell)}$ in the definitions of q_R , x_j , and the posterior quantities. This gives a practical coordinate-ascent scheme: update the single-effect factors one at a time, update fitted effects and residuals, and periodically update (ν_0, σ_0^2) by empirical Bayes.

4 Other models

4.1 Model C

$$\begin{cases} x_1 \mid \beta, \gamma = 1 \sim \mathcal{N}\left(\beta, \frac{1}{N}\right), \\ x_{-1} \mid x_1, z_1, \gamma = 1 \sim \mathcal{N}_{J-1}\left(x_1 z_1, \frac{1}{N} \bar{S}_1\right), \\ z_1 \mid S_1 \sim \mathcal{N}\left(\mu_1, \frac{S_1}{\nu_0}\right), \\ S_1 \sim \mathcal{IW}(\nu_0 \bar{S}_1, \nu_0 + J + 1), \\ \beta \sim \mathcal{N}(0, \sigma_0^2). \end{cases} \quad (26)$$

For a general causal SNP j , first integrate out S_j from the prior on z_j . Since

$$z_j \mid S_j \sim \mathcal{N}_{J-1}\left(\mu_j, \frac{S_j}{\nu_0}\right), \quad S_j \sim \mathcal{IW}(\nu_0 \bar{S}_j, \nu_0 + J + 1),$$

we have

$$z_j \sim t_{J-1, \nu_0+3}\left(\mu_j, \frac{1}{\nu_0+3} \bar{S}_j\right).$$

Equivalently, with the shape–rate convention,

$$\lambda_j \sim \text{Gamma}\left(\frac{\nu_0+3}{2}, \frac{1}{2}\right), \quad z_j \mid \lambda_j \sim \mathcal{N}_{J-1}\left(\mu_j, \frac{1}{\lambda_j} \bar{S}_j\right).$$

Therefore, conditional on x_j and λ_j ,

$$x_{-j} \mid x_j, \lambda_j, \gamma = j \sim \mathcal{N}_{J-1}\left(x_j \mu_j, \left\{\frac{1}{N} + \frac{x_j^2}{\lambda_j}\right\} \bar{S}_j\right).$$

Let

$$V_\beta = \sigma_0^2 + \frac{1}{N}, \quad q_R = x^\top \bar{R}^{-1} x, \quad r_j^2 = q_R - x_j^2,$$

where the last equality uses the block inverse identity

$$r_j^2 = (x_{-j} - x_j \mu_j)^\top \bar{S}_j^{-1} (x_{-j} - x_j \mu_j).$$

Then

$$\begin{aligned} p(x \mid \gamma = j) &= \mathcal{N}(x_j; 0, V_\beta) (2\pi)^{-(J-1)/2} |\bar{S}_j|^{-1/2} \\ &\times \int_0^\infty \left\{ \frac{1}{N} + \frac{x_j^2}{\lambda} \right\}^{-(J-1)/2} \\ &\times \exp \left[-\frac{r_j^2}{2 \left\{ 1/N + x_j^2/\lambda \right\}} \right] \\ &\times \frac{(1/2)^{(\nu_0+3)/2}}{\Gamma((\nu_0+3)/2)} \lambda^{(\nu_0+3)/2-1} \exp \left\{ -\frac{\lambda}{2} \right\} d\lambda. \end{aligned} \quad (27)$$

Since $|\bar{S}_j| = |\bar{R}|$ is common across SNPs, the PIP weights are

$$\begin{aligned}
w_j^C &= \mathcal{N}(x_j; 0, V_\beta) \int_0^\infty \left\{ \frac{1}{N} + \frac{x_j^2}{\lambda} \right\}^{-(J-1)/2} \\
&\quad \times \exp \left[-\frac{q_R - x_j^2}{2 \left\{ 1/N + x_j^2/\lambda \right\}} \right] \lambda^{(\nu_0+3)/2-1} \exp \left\{ -\frac{\lambda}{2} \right\} d\lambda, \\
\pi_j^C &= \frac{w_j^C}{\sum_{k=1}^J w_k^C}.
\end{aligned} \tag{28}$$

Thus model C has a one-dimensional integral per SNP. It does not preserve the exact null-SNP irrelevance property: the common statistic $q_R = x^\top \bar{R}^{-1} x$ appears inside an x_j -dependent integral, so it cannot be factored out and cancelled in the PIP normalization.